

# A strongly normalizing pure type system with ranks as sorts and Type-in-Type

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October 8, 2026

## Abstract

We present a dependently typed pure type system which contains a circular axiom  $s : s$  for a certain sort  $s$  and is nonetheless strongly normalizing. Girard’s paradox is avoided because our sorts are not closed under a  $\Pi$ -former: function types are placed in sorts according to their *rank*. A normalization proof is surprisingly simple (modulo general metatheoretical properties of pure type systems) and has been fully formalized in vanilla Rocq (formerly Coq) without any additional axioms.

## 1 Introduction

## 2 Pure type systems and Girard’s paradox

## 3 PTS-rk and its strong normalization

## 4 Formalizing PTS-rk in Rocq

Inductive characterization of the set of strongly normalizing terms follows [1].

## References

- [1] Felix Joachimski and Ralph Matthes. “Short proofs of normalization for the simply-typed  $\lambda$ -calculus, permutative conversions and Gödel’s T”. In: *Archive for Mathematical Logic* 42.1 (2003), pp. 59–87. DOI: 10.1007/s00153-002-0156-9. URL: <https://doi.org/10.1007/s00153-002-0156-9>.